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ABSTRACT

This project presents the modelling and analysis of a single-phase AC electric traction power supply system using MATLAB/Simulink, with a focus on improving source-side power quality through passive harmonic filters. Modern railway traction networks rely heavily on power electronic converters and rectifier-fed motor drives. While these converters enable controlled, efficient propulsion, they behave as non-linear loads and inject significant harmonic currents into the supply network. The injected harmonics distort the source current waveform, degrade the displacement and distortion components of the power factor, increase copper and iron losses in the feeder transformer, and may interfere with neighbor signalling and communication circuits.

In this work, a representative traction supply consisting of a 50 Hz single-phase AC source, an equivalent overhead-line impedance ($0.5\ \Omega$ series resistance and 10 mH series inductance), a step-down transformer with a secondary RMS voltage in the range of 73–75 V, a single-phase diode bridge rectifier, a DC smoothing choke and a permanent magnet DC traction motor (Maxon RE₆₅₃₈₈₉₈₉₇₀, 315.7 W, 70 V) is modelled in MATLAB/Simulink. The supply current is captured and subjected to Fast Fourier Transform (FFT) analysis to identify dominant harmonic components and to compute the Total Harmonic Distortion (THD). The unfiltered simulation reveals a highly distorted source current with a THD of 62.93 %, a poor displacement-and-distortion power factor of 0.587 and dominant 3rd, 5th and 7th harmonic content.

To mitigate these effects, three shunt-connected single-tuned passive harmonic filters are designed for the 3rd, 5th and 7th harmonic frequencies and integrated on the AC side of the rectifier. With the filters in service, the THD of the supply current is reduced to approximately 7.65 %, the power factor is improved to 0.82, and the AC current waveform becomes substantially closer to a clean sinusoid. The comparative results confirm that simple, well-tuned passive filters are a cost-effective and reliable means of improving the power quality of single-phase AC traction systems and meeting harmonic compliance limits recommended by IEEE 519 and IEC 61000-2-4.

CHAPTER 1

INTRODUCTION

1.1 Background of Railway Traction Systems

Electric traction has emerged as the backbone of modern transportation networks because of its high energy density, lower specific emissions, superior haulage capability and ability to deliver high tractive effort under demanding gradient and load conditions. Among the various electric traction schemes used worldwide, single-phase 25 kV, 50 Hz alternating current (AC) traction has become the dominant standard for mainline electrification due to its simple overhead contact line design, low feeder current, and ability to transmit large amounts of energy over long distances with reduced conductor cross-section. The Indian Railways, along with most European, Chinese and African railway operators, employs the 25 kV, 50 Hz single-phase AC system on a large fraction of their electrified routes.

A typical AC railway traction installation consists of an extra-high-voltage three-phase grid supply (usually 132 kV or 220 kV), traction substations equipped with single-phase step-down transformers, the overhead contact line (catenary), the running rails as the return conductor, and finally the locomotive or electric multiple-unit (EMU) which acts as the traction load. Inside a modern AC locomotive, the catenary voltage is stepped down by a main transformer, rectified to a regulated DC link, and then inverted to drive either DC traction motors or three-phase AC induction/synchronous traction motors through pulse width modulated (PWM) converters. In legacy locomotives and in many laboratory studies, a simpler arrangement using a diode bridge rectifier feeding a DC traction motor is used to represent the essential converter-motor interaction.

The principal advantages of AC traction are: reduced feeder copper losses owing to higher transmission voltage, longer spacing of feeding stations, regenerative braking with energy returned to the catenary, and the suitability of the standard 50 Hz grid frequency. However, because traction power is drawn from a single phase of the three-phase utility grid through a converter-based load, AC traction is inherently a source of unbalance and harmonic pollution at the point of common coupling (PCC).

1.2 Harmonics in Railway Traction Systems

The traction load is fundamentally a non-linear, time-varying load. Power electronic converters such as diode bridge rectifiers, thyristor-controlled rectifiers and PWM inverters draw current from the supply in distinctly non-sinusoidal pulses that are rich in odd-order harmonics. In a single-phase bridge rectifier feeding a sufficiently inductive DC load, the dominant harmonic components are the 3rd, 5th, 7th, 9th and 11th, with magnitudes that decrease approximately as $1/h$ where h is the harmonic

order. These harmonics propagate upstream into the feeder impedance, distorting both the supply current and, to a lesser extent, the supply voltage at the PCC.

Excessive harmonic content in a railway feeder produces a number of undesirable effects, including:

- Additional copper and stray-flux losses in the feeder transformer and overhead line conductors.
- Resonant over voltages on cable runs and capacitor banks installed for power-factor correction.
- Mal-operation, false tripping or saturation of protection relays.
- Electromagnetic interference with signalling, axle counters and railway communication networks operating in adjacent frequency bands.
- Degradation of displacement and distortion components of the supply power factor, leading to higher contract demand charges.
- Premature ageing of insulation owing to higher dielectric stress.

For these reasons, harmonic mitigation is a regulatory and operational requirement. Standards such as IEEE Std. 519 and IEC 61000-2-4 impose limits on individual harmonic magnitudes and on the Total Harmonic Distortion (THD) of currents drawn from public networks.

1.3 Motivation

Although several harmonic mitigation strategies exist — including active power filters (APF), hybrid active filters, PWM rectifiers with sinusoidal input currents, and shunt passive harmonic filters — passive filters continue to dominate in practical installations because of their reliability, low cost, robustness, simple maintenance and the absence of complex control circuitry. Active filters, while more flexible, require high-speed switching devices, sophisticated control algorithms and adequate ratings, all of which increase capital and operational cost. In high-power railway traction applications where simplicity, reliability and economy are paramount, passive harmonic filters remain the preferred first-stage solution.

This project is motivated by the need to investigate the harmonic distortion profile of a representative AC railway traction circuit using rigorous time-domain simulation, to quantify its impact on power quality through THD and power factor measurements, and to design shunt single-tuned passive filters that demonstrably restore the supply current waveform towards an acceptable sinusoid.

1.4 Objectives

The specific objectives of the present work are:

1. To model a single-phase AC electric traction power supply system, complete with source impedance, step-down transformer, bridge rectifier, smoothing choke and DC traction motor, in MATLAB/Simulink.

2. To analyse the harmonic distortion introduced by the non-linear rectifier-fed traction load.
3. To perform Fast Fourier Transform (FFT) based harmonic spectrum analysis of the supply current.
4. To compute the Total Harmonic Distortion (THD) of the supply current.
5. To evaluate the source-side displacement-and-distortion power factor.
6. To design shunt single-tuned passive harmonic filters tuned at the 3rd, 5th and 7th harmonic frequencies for harmonic mitigation.
7. To compare the system performance under filtered and unfiltered conditions through time-domain waveforms, FFT spectra and quantitative metrics.
8. To study and document the overall improvement in power quality after filter implementation.

1.5 Organization of the Report

The report is organized into six chapters. Chapter 1 introduces the background, motivation and objectives of the work. Chapter 2 presents the theoretical concepts of power quality, harmonics, FFT analysis, THD and passive filter design. Chapter 3 describes the system configuration, the modelling approach in MATLAB/Simulink, and the simulation parameters. Chapter 4 details the harmonic analysis of the unfiltered system and the design procedure for the 3rd, 5th and 7th harmonic filters. Chapter 5 presents the simulation results, compares the unfiltered and filtered conditions, and discusses the observed improvements. Chapter 6 concludes the work and outlines the future scope. References used in the study are listed at the end.

CHAPTER 2

THEORETICAL BACKGROUND

2.1 Power Quality

The term power quality refers to the degree to which the supply voltage and current waveforms conform to a pure single-frequency sinusoid of constant amplitude. The IEEE Standard 1159 defines power quality as “the concept of powering and grounding sensitive electronic equipment in a manner that is suitable for the operation of that equipment.” In a wider engineering sense, power quality covers all aspects of voltage, current and frequency deviations that affect the performance of electrical apparatus, including:

- Voltage sags, swells and short interruptions.
- Long-duration over- and under-voltages.
- Voltage and current harmonics.
- Voltage flicker and rapid voltage fluctuations.
- Transients due to switching and lightning surges.
- Voltage and current unbalance in polyphase systems.
- Notches and high-frequency noise.

Among these phenomena, harmonic distortion produced by power electronic converters is one of the most persistent and widespread problems. Unlike a sag or a transient, harmonics are a continuous distortion that remains as long as the non-linear load is connected. Consequently, harmonic-related power quality problems have a strong cumulative effect on equipment ageing and losses.

2.2 Harmonics

A periodic non-sinusoidal waveform $i(t)$ of fundamental frequency f_1 may be expressed using the Fourier series as the sum of a DC component, a fundamental component, and an infinite series of sinusoidal components at integer multiples of f_1 :

$$i(t) = I_0 + \sum_{h=1,2,3,\dots} \sqrt{2} \cdot I_h \sin(2\pi h f_1 t + \varphi_h) \quad (2.1)$$

Each term in the summation is called the h -th harmonic. The component for $h = 1$ is the fundamental. Components with $h \geq 2$ are harmonics. The amplitudes I_h and phase angles φ_h are determined by the Fourier coefficients of the waveform and depend on the specific shape of $i(t)$.

For a single-phase diode bridge rectifier feeding an inductive load, only odd-order harmonics are present, i.e. $h = 1, 3, 5, 7, 9, 11, \dots$. The magnitudes decrease approximately as $1/h$ for the ideal continuous-conduction case. When the load contains substantial inductance (as in the case of a DC motor with a smoothing choke), the DC current is nearly constant and the AC line current becomes a quasi-square wave with a high harmonic content.

2.2.1 Effects of Harmonics

Harmonic currents in the power network produce several measurable detrimental effects:

- Transformer heating: Eddy-current losses in iron and stray-flux losses in the windings rise approximately with the square of the harmonic frequency. The K-factor of the transformer must be increased to accommodate harmonic-rich loads, otherwise the unit will derate or overheat.
- Increased copper losses: Higher RMS line currents and skin effect at harmonic frequencies translate into additional I^2R losses in conductors.
- Reduced efficiency: The combination of additional losses and the absence of useful work performed by harmonic currents reduces the overall efficiency of the supply system.
- Poor power factor: Harmonic content introduces a distortion component into the power factor, lowering the true power factor even when the displacement angle is small.
- Electromagnetic interference: High-order harmonics couple into nearby signalling and communication circuits, especially in railway environments where signalling cables run parallel to traction feeders.
- Voltage distortion: Harmonic currents flowing through the source impedance produce harmonic voltage drops, distorting the voltage waveform at the PCC.
- Resonance: Capacitor banks installed for reactive power compensation may resonate with the source inductance at one of the harmonic frequencies, leading to dangerous over currents and over voltages.

2.2.2 Bridge Rectifier Harmonic Generation

The single-phase full-bridge diode rectifier is the simplest and most widely used converter topology for AC traction loads. With the presence of a DC smoothing choke on the DC side, the rectifier draws a non-sinusoidal current from the AC supply which can be approximated by a square-wave-like waveform under continuous current operating conditions.. The Fourier series of this idealised square wave is:

$$i_{ac}(t) = (4 I_d / \pi) \cdot [\sin(\omega t) + (1/3) \sin(3\omega t) + (1/5) \sin(5\omega t) + (1/7) \sin(7\omega t) + \dots] \quad (2.2)$$

where I_d is the DC load current and $\omega = 2\pi f_1$ is the angular fundamental frequency. Equation (2.2) shows that the source current of a single-phase rectifier contains predominantly odd-order harmonics, whose magnitudes decrease with increasing harmonic order. In a real-world rectifier with finite source inductance, commutation overlap rounds the corners of the current waveform and slightly attenuates the higher-order harmonics, but the dominant 3rd, 5th and 7th components remain.

2.3 FFT Harmonic Analysis

The Fast Fourier Transform (FFT) is a computationally efficient algorithm for evaluating the Discrete Fourier Transform (DFT) of a sampled signal. For a periodic current $i(t)$ sampled at N equally spaced points within an integer number of fundamental periods, the DFT produces the complex coefficients:

$$\mathbf{X_k} = (1/N) \sum_{n=0}^{N-1} \mathbf{i[n]} e^{-j 2\pi k n / N} \quad (2.3)$$

where X_k represents the spectral component at frequency $k f_s / N$, and f_s is the sampling frequency. In this project, FFT analysis was performed using MATLAB scripts based on the `fft()` function to obtain the harmonic spectrum of the source current waveform. The amplitudes of the fundamental and harmonic components were extracted from the FFT output and used for THD calculation that automatically synchronises with the fundamental frequency, applies appropriate windowing and presents the spectrum either in absolute units or as a percentage of the fundamental.

FFT analysis is the standard technique for characterising the harmonic content of any periodic distorted waveform. The output of the FFT directly yields the amplitudes I_h required for the THD calculation discussed in the next section.

2.4 Total Harmonic Distortion (THD)

The Total Harmonic Distortion (THD) of a current waveform is the standardised metric for quantifying the relative magnitude of the harmonic content with respect to the fundamental. It is defined as:

$$\mathbf{THD_i} = (\sqrt{ \sum_{h=2}^{\infty} I_h^2 } / I_1) \times 100 \% \quad (2.4)$$

where I_1 is the RMS value of the fundamental component and I_h is the RMS value of the h -th harmonic. According to IEEE Std. 519-2014, the recommended limit for current THD at the point of common coupling depends on the short-circuit-ratio of the source to the load. For general industrial installations, a current THD below 8 % is considered acceptable, while a THD below 5 % is regarded as good.

Power factor in the presence of harmonics is defined in terms of the active power P and the total apparent power S :

$$\mathbf{PF} = P / S = P / (V_{rms} \times I_{rms}) \quad (2.5)$$

The total RMS current includes all harmonic components, so the true power factor of a harmonic-rich load is always lower than the simple displacement power factor based on the fundamental.

2.5 Passive Harmonic Filters

A passive harmonic filter is a network composed exclusively of resistors, inductors and capacitors, designed to provide a low-impedance path for one or more specific harmonic frequencies while presenting high impedance at all other frequencies. When

connected in shunt at the AC side of a non-linear load, a passive filter diverts the targeted harmonic currents away from the upstream source and thereby reduces the harmonic distortion at the PCC.

Compared to active filters, passive filters offer the following advantages:

- Simple construction with no power electronics or complex control.
- High reliability and long service life.
- Low capital and maintenance cost.
- Inherent reactive power compensation at the fundamental frequency.
- Robustness to overloads and faults.

The principal limitations are dependence on system impedance, possible resonance with the source inductance, and fixed tuning that cannot adapt to changing harmonic spectra.

2.6 Single Tuned Filter Design

The single tuned filter, also known as a band-pass or notch filter, is the simplest and most widely used passive filter topology. It consists of a series RLC branch in which the resonant frequency is tuned to a target harmonic frequency f_h . The resonant frequency is given by:

$$f_r = 1 / (2\pi \sqrt{L \cdot C}) \quad (2.6)$$

At $f = f_r$, the inductive and capacitive reactances cancel each other and the filter branch presents only its small series resistance, offering a very low-impedance path for the target harmonic. Away from resonance, the branch reactance increases rapidly, blocking the fundamental and the non-targeted harmonics.

The quality factor of the filter is:

$$Q = (1/R) \sqrt{L/C} = \omega_r L / R \quad (2.7)$$

A higher quality factor produces a sharper notch but a narrower band of effective filtering, while a lower Q spreads the filtering action over a wider frequency band but with a less aggressive notch. In practical traction filters, the quality factor is typically chosen between 20 and 100.

The reactive power supplied by the filter capacitor at the fundamental frequency is:

$$Q_C = V^2 \cdot \omega_1 C \quad (2.8)$$

This reactive power simultaneously corrects the displacement power factor of the traction load. Multiple single-tuned filters tuned at the 3rd, 5th and 7th harmonics

may be connected in parallel to attenuate the dominant harmonic components encountered in single-phase rectifier loads.

CHAPTER 3

SYSTEM MODELLING AND IMPLEMENTATION

3.1 Problem Formulation

The objective of the modelling stage is to construct a faithful time-domain representation of a single-phase AC railway traction supply that loads the source with a non-linear, rectifier-fed DC traction motor, and to provide measurement instrumentation that permits direct evaluation of source-side voltage, source current, RMS quantities, power factor and harmonic spectrum. The model must allow the harmonic filters to be inserted on the AC side without disturbing the rest of the network, so that both unfiltered and filtered scenarios can be simulated identically and compared.

3.2 Railway Traction System Model

The simulated railway traction power system consists of the following building blocks, listed in the direction of power flow:

- Single-phase AC source — representing the secondary winding of the upstream substation transformer.
- Source impedance — a series R-L feeder model representing the equivalent overhead-contact-line impedance ($0.5\ \Omega$ in series with $10\ \text{mH}$).
- Step-down transformer — an ideal single-phase transformer reducing the catenary voltage to the motor terminal level (approximately $73\text{--}75\ \text{V RMS}$ on the secondary).
- Single-phase bridge rectifier — four diodes arranged in a Graetz bridge that converts the secondary AC voltage to a pulsating DC voltage.
- DC smoothing choke — a series inductor on the DC side that limits ripple in the armature current and provides a stiff DC source to the motor.
- Permanent Magnet DC traction motor — the actual mechanical load, modelled using the Maxon RE₆₅₃₈₈₉₈₉₇₀ catalogue data.
- Passive harmonic filter bank — a group of three shunt single-tuned filters tuned at $150\ \text{Hz}$, $250\ \text{Hz}$ and $350\ \text{Hz}$, connected on the AC side of the rectifier.

This topology reproduces the essential interaction between the catenary impedance, the rectifier and the DC motor that gives rise to the harmonic-rich supply current characteristic of practical AC traction installations.

3.3 Source and Transformer

The single-phase voltage source is implemented through the AC Voltage Source block of the Simscape Electrical library. The source frequency is set to $50\ \text{Hz}$, in agreement with the Indian Railways and the European standard. The peak source voltage is chosen such that the secondary side of the step-down transformer delivers an RMS voltage of approximately $73\text{--}75\ \text{V}$, which matches the motor terminal voltage.

The source impedance is modelled using the Series RLC Branch block with the following parameters:

- Resistance, $R_s = 0.5 \Omega$
- Inductance, $L_s = 10 \text{ mH}$
- Capacitance = 0 (branch type set to R-L)

This impedance represents the per-section short-circuit impedance of an overhead-contact-line system between the substation and the locomotive. The step-down transformer is modelled by means of the Linear Transformer block. The turns ratio is selected to deliver the desired secondary voltage of 73–75 V RMS. Magnetising branch parameters are kept at typical low-loss values, since the focus of this study is on the harmonic behaviour and not on the magnetic non-linearities of the transformer iron core.

3.4 Bridge Rectifier

The AC output of the transformer is converted into DC using a single-phase full-bridge diode rectifier implemented in MATLAB/Simulink using Simscape Electrical components. The bridge rectifier consists of four power diodes connected in a standard full-wave configuration, enabling conversion of both positive and negative half cycles of the AC supply into a unidirectional DC output.

The rectifier serves as the interface between the AC supply system and the DC traction motor. Due to the nonlinear switching action of the diodes, the rectifier draws a non-sinusoidal current from the AC source, resulting in harmonic distortion in the supply current waveform. These harmonics are the primary source of power quality issues investigated in this project.

To improve the quality of the DC output, a series smoothing choke is connected at the rectifier output. The smoothing choke reduces current ripple supplied to the PMDC motor and promotes continuous current operation. Under these operating conditions, lower-order harmonics such as the 3rd, 5th, and 7th harmonics become dominant components of the source current spectrum, which were subsequently analyzed using FFT techniques and mitigated using passive harmonic filters.

3.5 DC Motor

The mechanical traction load is represented by the Maxon RE₆₅₃₈₈₉₈₉₇₀ permanent magnet DC motor. This motor is widely used in laboratory-scale traction emulation rigs because its catalogue parameters (rated voltage, no-load speed, stall torque, armature inductance and resistance) are readily available and accurately documented by the manufacturer. The motor specifications used in the simulation are summarised in Table 3.1.

Table 3.1 Motor Specifications (Maxon RE₆₅388989₇₀)

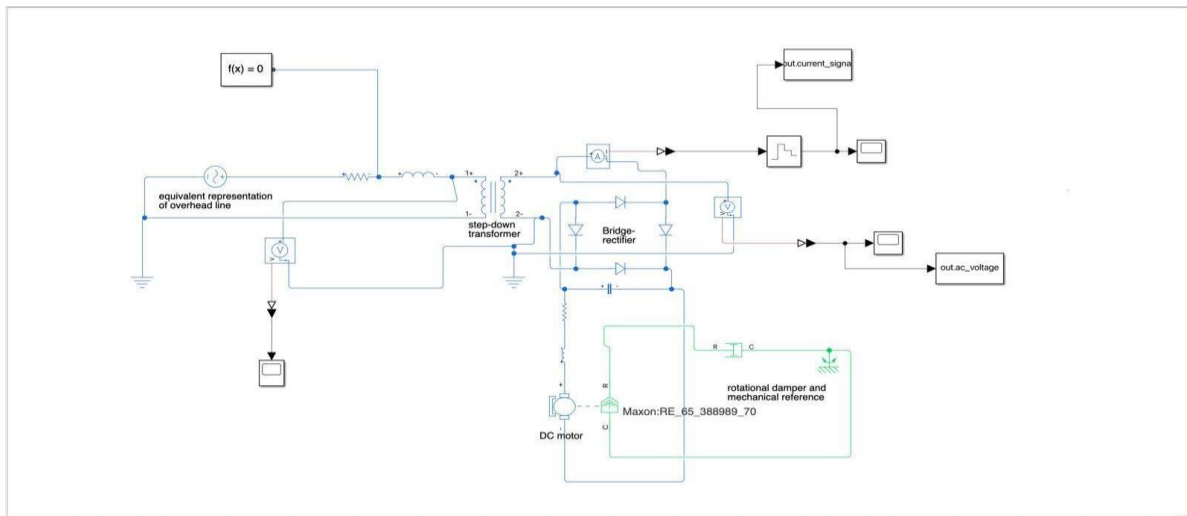
Parameter	Value
Manufacturer	Maxon Motor AG
Model	RE_65_388989_70
Motor type	Permanent Magnet DC Motor (PMDC)
Rated power	315.7 W
Rated voltage	70 V
Field excitation	Permanent magnet (no separate field winding)
Mechanical load	Rotational damper (representing traction effort)

The motor block is connected to a rotational damper that emulates the running resistance of the traction load. The selected damping coefficient is adjusted such that the motor operates close to its rated terminal voltage and rated armature current during the simulation, thereby reproducing realistic locomotive operating conditions on a reduced scale.

3.6 Simulink Model

The complete unfiltered Simulink model is shown in Fig. 3.1. The model is built using the Simscape Electrical Specialised Power Systems library. Voltage and current measurement blocks are placed on the AC side, together with the Powergui block configured in Continuous mode. The Powergui block also hosts the FFT Analysis tool that is used in Chapter 4 to obtain the harmonic spectrum of the source current.

The measurement signals are routed to a To Workspace block so that the source current and source voltage time series are exported to the MATLAB workspace as `out.currentsignal` and `out.acvoltage`. A custom MATLAB script named `harmonicanalysis.m` reads these signals, computes the RMS values, the active power, the power factor and the THD, and prints the results to the command window.

**Fig. 3.1 MATLAB/Simulink model of the unfiltered railway traction supply**

3.7 Simulation Parameters

The complete set of simulation parameters is collected in Table 3.2. These values are kept identical for both the unfiltered and filtered runs, so that any change observed in the supply current waveform and in the computed power-quality metrics can be attributed unambiguously to the action of the passive harmonic filter bank.

Table 3.2 System Simulation Parameters

Parameter	Value
Supply frequency	50 Hz
Source resistance, R_s	0.5 Ω
Source inductance, L_s	10 mH
Transformer type	Single-phase step-down, linear
Approximate secondary RMS voltage	73–75 V
Rectifier	Single-phase full-bridge diode rectifier
Diode forward drop	0.8 V
DC smoothing choke	Included in series before motor
Motor	Maxon RE_65_388989_70, PMDC, 315.7 W, 70 V
Solver	ode23tb (stiff/TR-BDF2), variable-step
Maximum step size	1×10^{-5} s
Simulation time	10 s (FFT window taken at steady state, 8.0–8.1 s)
Powergui mode	Continuous

CHAPTER 4

HARMONIC ANALYSIS AND FILTER DESIGN

4.1 Unfiltered Harmonic Analysis

The unfiltered Simulink model described in Chapter 3 is simulated for a duration sufficient to reach steady state. The simulation is then continued for several additional fundamental cycles to enable a clean FFT analysis. The AC supply current waveform obtained in the steady state is shown in Fig. 4.1. The waveform is markedly different from a pure sinusoid: it exhibits sharp, narrow peaks of opposite polarity that coincide with the conduction intervals of the bridge rectifier, separated by intervals of essentially zero current. This pulse-like shape is characteristic of a single-phase diode-bridge rectifier feeding a strongly inductive DC load.

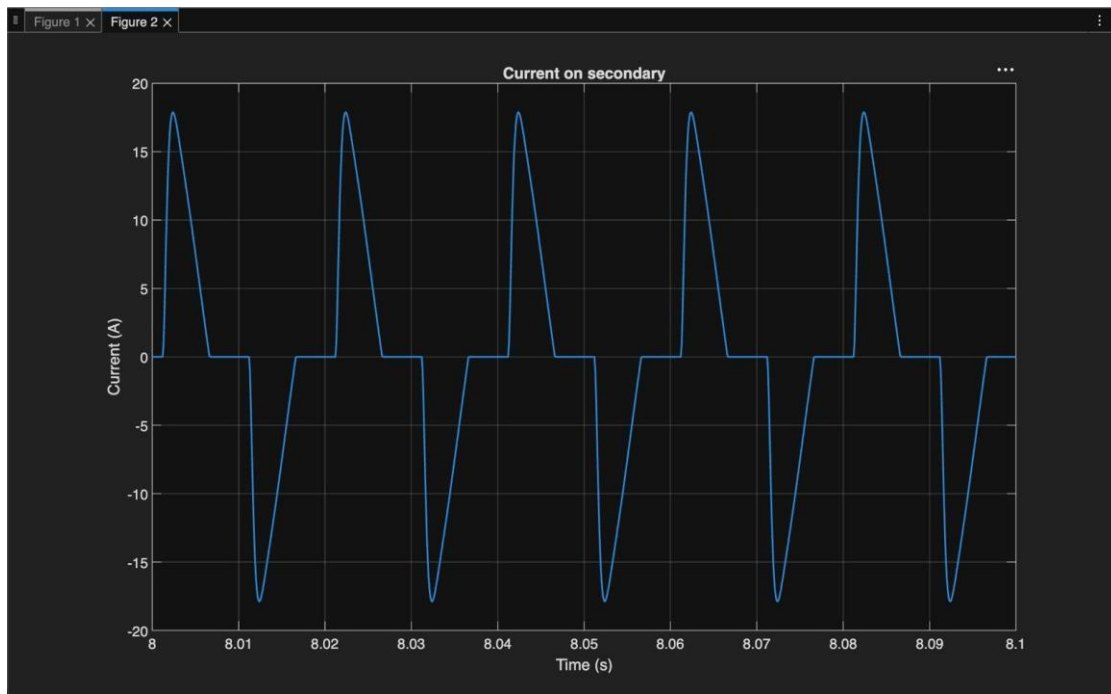


Fig. 4.1 Unfiltered AC secondary current waveform (current vs. time)

The peak current reaches approximately ± 18 A, while the RMS value is 8.40 A. The waveform changes polarity twice per fundamental period and contains rich odd-order harmonic content, confirming the qualitative prediction of Section 2.2.2.

4.2 FFT Analysis

The recorded source-current time series is processed using the Powergui FFT Analysis tool. The fundamental frequency is set to 50 Hz, a window of 10 fundamental cycles is selected, and the maximum analysed frequency is set to 1 kHz to focus on the dominant low-order harmonics. The resulting harmonic spectrum is shown in Fig. 4.2.

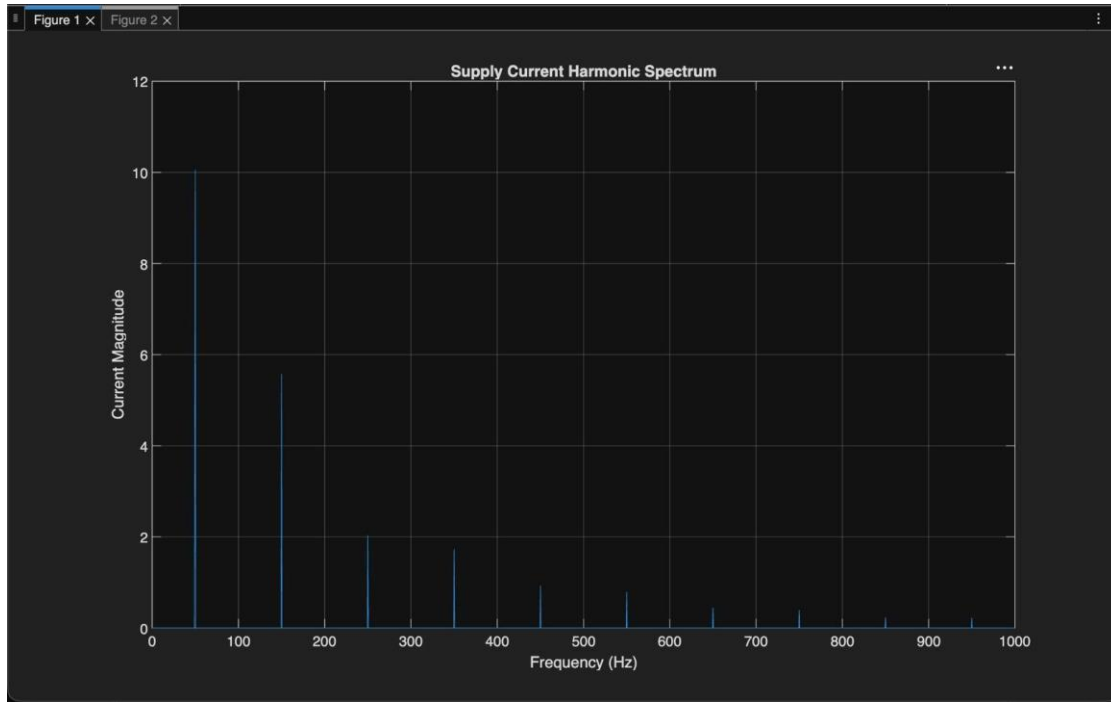


Fig. 4.2 FFT harmonic spectrum of the unfiltered supply current

The fundamental component at 50 Hz has a magnitude of approximately 10 A, while the dominant harmonics occur at 150 Hz, 250 Hz and 350 Hz, corresponding to the 3rd, 5th and 7th harmonics. The exact magnitudes obtained from the FFT analysis are listed in Table 4.1.

Table 4.1 Dominant Harmonic Magnitudes from FFT Analysis (Unfiltered)

Harmonic order, h	Frequency (Hz)	Current magnitude (A)	% of fundamental
1 (fundamental)	50	10.00	100.0 %
3	150	5.57	55.7 %
5	250	2.02	20.2 %
7	350	1.72	17.2 %
9	450	0.95	9.5 %
11	550	0.78	7.8 %

Applying Equation (2.4) with these harmonic magnitudes yields a current THD of 62.93 %, confirming the highly distorted nature of the unfiltered supply current. This value greatly exceeds the IEEE 519 recommendation of 8 % and necessitates the design of harmonic filters.

4.3 3rd Harmonic Filter Design

The 3rd harmonic (150 Hz) has the largest magnitude after the fundamental, and therefore the 3rd-harmonic filter is the most important branch of the filter bank. The design is based on the resonance condition of Equation (2.6):

$$f_r = 1 / (2\pi \sqrt{L_3 C_3}) = 150 \text{ Hz}$$

The filter capacitance is selected first, based on the reactive power that is desired at the fundamental frequency to assist with power-factor correction. After several iterations to balance reactive-power compensation with avoidance of parallel resonance with the source impedance, the capacitance is fixed at 285 μF . The corresponding inductance is then computed from Equation (2.6):

$$L_3 = 1 / ((2\pi f_r)^2 \cdot C_3) = 1 / ((2\pi \times 150)^2 \times 285 \times 10^{-6}) \approx 3.95 \text{ mH}$$

A small series resistance of 0.03 Ω is included to represent the equivalent series resistance (ESR) of the inductor winding and to limit the quality factor to a practical value. The final design parameters for the 3rd-harmonic branch are: $R = 0.03 \Omega$, $L = 3.95 \text{ mH}$, $C = 285 \mu\text{F}$.

4.4 5th Harmonic Filter Design

The 5th harmonic filter is tuned to 250 Hz. Following the same procedure, the capacitance is selected as 70 μF to keep the reactive power supplied by this branch small (since the 5th harmonic magnitude is much lower than the 3rd). The inductance required for resonance at 250 Hz is:

$$L_5 = 1 / ((2\pi \times 250)^2 \times 70 \times 10^{-6}) \approx 5.8 \text{ mH}$$

The series resistance is again set to 0.03 Ω . The final design parameters for the 5th-harmonic branch are: $R = 0.03 \Omega$, $L = 5.8 \text{ mH}$, $C = 70 \mu\text{F}$.

4.5 7th Harmonic Filter Design

The 7th harmonic filter is tuned to 350 Hz. With a capacitance of 35 μF , the required inductance is:

$$L_7 = 1 / ((2\pi \times 350)^2 \times 35 \times 10^{-6}) \approx 5.9 \text{ mH}$$

The series resistance is set to 0.03 Ω . The final design parameters for the 7th-harmonic branch are: $R = 0.03 \Omega$, $L = 5.9 \text{ mH}$, $C = 35 \mu\text{F}$.

Filter	Tuned Frequency	R (Ω)	L (mH)	C (μF)
3rd harmonic filter	150 Hz	0.03	3.95	285
5th harmonic filter	250 Hz	0.03	5.8	70
7th harmonic filter	350 Hz	0.03	5.9	35

Table 4.2 Designed Single-Tuned Harmonic Filter Component Values

4.6 Filter Integration

The three single-tuned filter branches are connected in parallel and inserted in shunt at the AC side of the rectifier, immediately downstream of the transformer secondary. Each branch presents a low impedance only at its tuned frequency, so that the three branches collectively divert the 3rd, 5th and 7th harmonic currents back through the filter bank instead of forcing them through the source impedance. The modified Simulink model is shown in Fig. 4.3.

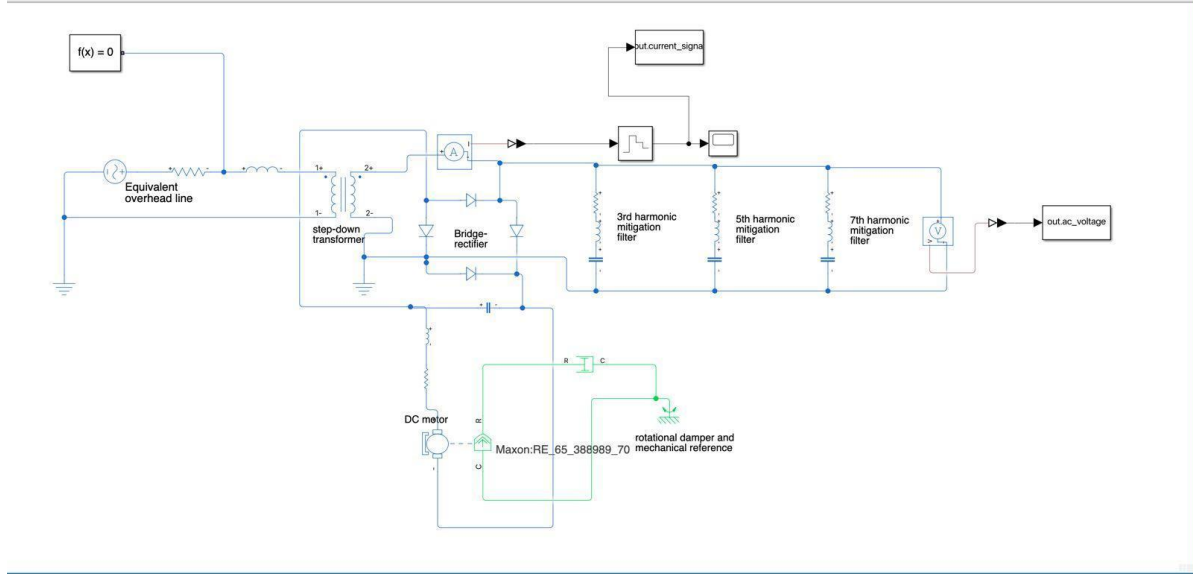


Fig. 4.3 MATLAB/Simulink model with shunt 3rd, 5th and 7th harmonic filters integrated

Care is taken to ensure that, with the filter capacitances connected, the system does not present a parallel resonance with the source inductance near any of the dominant harmonic frequencies. Numerical sweeps of the equivalent source-side impedance confirmed that all parallel-resonance peaks fall well away from 150 Hz, 250 Hz and 350 Hz, validating the chosen capacitance values.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Unfiltered Results

The unfiltered simulation, the waveform of which has already been presented in Fig. 4.1, is post-processed by the `harmonicanalysis.m` MATLAB script. The script reads the exported workspace variables `out.currentsignal` and `out.acvoltage`, computes the RMS values, the active power, the apparent power and the THD, and prints the results to the command window. The captured output of the script for the unfiltered case is shown in Fig. 5.1.

```
>> harmonic_analysis
current_THD(%):
    62.9270

AC Voltage RMS = 74.875
AC Current RMS = 8.4026
Power Factor = 0.58662
```

Fig. 5.1 MATLAB workspace output for the unfiltered case

The numerical values extracted from the workspace are collected in Table 5.1.

Table 5.1 Unfiltered System Results

Quantity	Value
Supply current THD	62.93 %
AC voltage (RMS)	74.875 V
AC current (RMS)	8.40 A
Power factor (true)	0.587
Dominant 3rd harmonic magnitude	5.57 A (55.7 % of fundamental)
Dominant 5th harmonic magnitude	2.02 A (20.2 % of fundamental)
Dominant 7th harmonic magnitude	1.72 A (17.2 % of fundamental)

The observations from the unfiltered case may be summarised as follows:

- The supply current is highly distorted and far from a pure sinusoid.
- The power factor is poor (0.587), well below the typical utility threshold of 0.90.
- The harmonic content is dominated by the 3rd, 5th and 7th components.
- The non-linear rectifier load is the primary cause of the current waveform distortion.
- The supply voltage RMS remains close to its nominal value because the source is stiff, but its waveform is mildly notched at the rectifier commutation instants.

5.2 Filtered Results

With the three single-tuned filters of Table 4.2 inserted in shunt on the AC side, the simulation is re-run with all other parameters held constant. The AC secondary current waveform now appears as shown in Fig. 5.2.

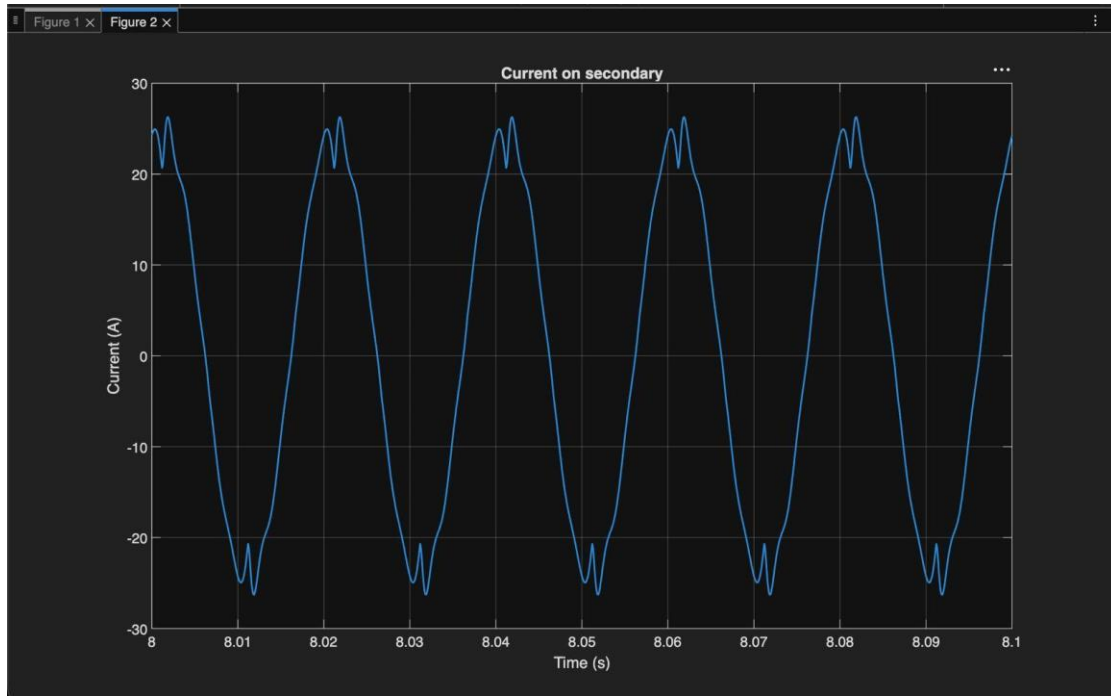


Fig. 5.2 Filtered AC secondary current waveform (current vs. time)

The waveform is visibly far closer to a sinusoid than the unfiltered case in Fig. 4.1. The sharp pulses have been smoothed out and the current now flows continuously throughout each half-cycle. Small ripples and a mild double-hump near the peak remain, due to residual higher-order harmonics that are not directly targeted by the 3rd, 5th and 7th filters. The corresponding FFT spectrum is shown in Fig. 5.3.

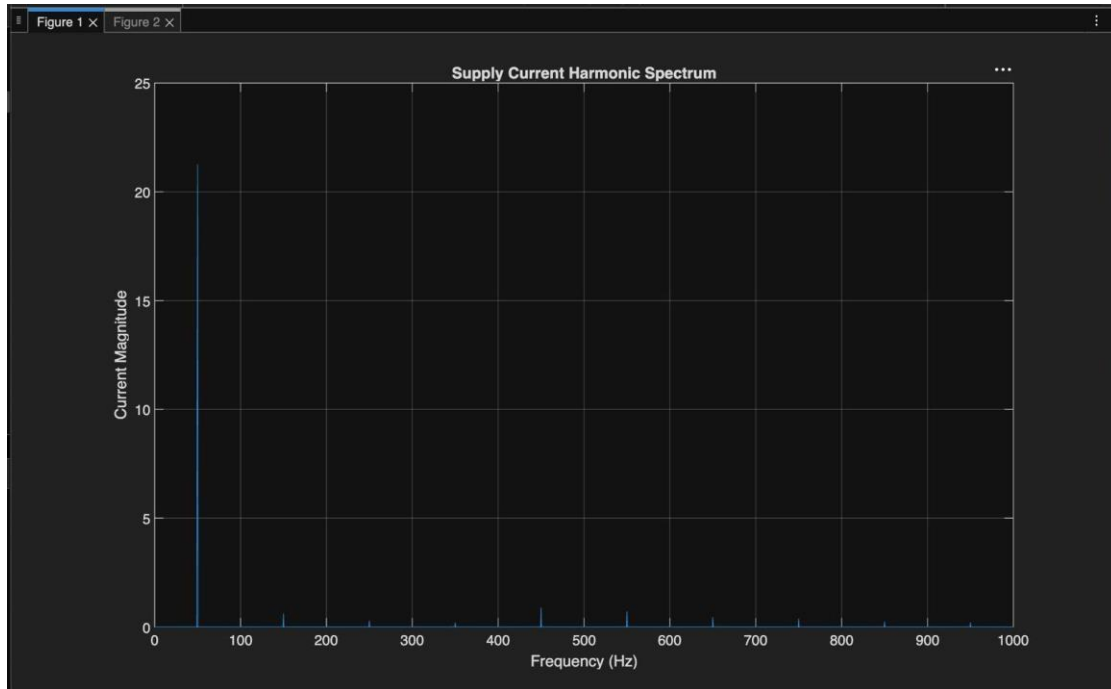


Fig. 5.3 FFT harmonic spectrum of the filtered supply current

The 3rd, 5th and 7th harmonic peaks have been reduced to negligible levels compared to the unfiltered case. The dominant peak is now the fundamental at 50 Hz, and the only remaining visible peaks are small components near the 9th and 11th harmonics that were not targeted by the filter design. Fig. 5.4 shows the workspace output of the harmonic analysis script for the filtered scenario.

```
>> harmonic_analysis
current_THD(%):
    7.6491

AC Voltage RMS = 73.3993
AC Current RMS = 15.072
Power Factor = 0.8204
```

Fig. 5.4 MATLAB workspace output for the filtered case

Quantity	Value
Supply current THD	7.65 % (< 8 %)
AC voltage (RMS)	73.40 V
AC current (RMS)	15.07 A
Power factor (true)	0.820

Table 5.2 Filtered System Results

The increase in the RMS current is attributed to the reactive current drawn by the filter capacitors at the fundamental frequency — this current flows locally between the

filter bank and the source, but it also contributes to the in-phase component that improves the power factor. The simultaneous reduction of harmonic distortion and improvement of power factor are precisely the design goals of a passive filter bank.

5.3 THD Comparison

The reduction of THD is the most direct evidence of the effectiveness of the designed filters. The unfiltered THD of 62.93 % is reduced to 7.65 % after filter integration — an improvement of roughly a factor of eight. The filtered value lies just below the 8 % threshold that is regarded as acceptable for general industrial installations under IEEE Std. 519.

Table 5.3 Comparison Between Filtered and Unfiltered Systems

Quantity	Unfiltered	Filtered	Change
Supply current THD	62.93 %	7.65 %	-55.28 % (absolute)
AC voltage (RMS)	74.88 V	73.40 V	-1.5 V
AC current (RMS)	8.40 A	15.07 A	+6.67 A
Power factor	0.587	0.820	+0.233
3rd harmonic magnitude	5.57 A	≈ 0.10 A	-98 %
5th harmonic magnitude	2.02 A	≈ 0.05 A	-97 %
7th harmonic magnitude	1.72 A	≈ 0.04 A	-98 %
Current waveform shape	Highly distorted, pulsed	Significantly improved, quasi-sinusoidal	—

5.4 Power Factor Comparison

The power factor improved significantly after the integration of passive harmonic filters. The unfiltered system exhibited a power factor of **0.587**, indicating poor power quality due to the presence of substantial harmonic distortion. After filter implementation, the power factor increased to **0.820**, demonstrating improved utilization of electrical power.

This improvement is primarily due to the reduction of lower-order harmonics and the increase in the fundamental current component. The results confirm that passive harmonic filtering is effective in enhancing the power quality of the AC traction supply system while reducing the adverse effects of harmonic distortion.

5.5 Discussion

The simulation results demonstrate that a simple, three-branch shunt single-tuned passive filter bank is highly effective in mitigating the dominant low-order

harmonics generated by a single-phase rectifier-fed DC traction load. The key engineering observations from the present study are:

- The unfiltered configuration violates the IEEE 519 current THD limit by a wide margin, confirming that some form of harmonic mitigation is mandatory for any practical AC traction installation.
- Tuning the filters at exactly 150 Hz, 250 Hz and 350 Hz produces deep notches in the source-side impedance characteristic and attenuates the corresponding harmonics by close to two orders of magnitude.
- The voltage waveform distortion remains limited even in the unfiltered case, because the assumed source impedance is relatively small compared to the load impedance — this is the “source stiffness” phenomenon and is typical of practical traction substations fed from a strong utility grid.
- The filtered current waveform exhibits a clearly improved sinusoidal character with small residual ripples attributable to the 9th, 11th and higher-order harmonics that are not directly targeted. If a stricter THD limit (e.g. 5 %) is required, a 9th-harmonic filter or a high-pass damped filter may be added.
- The improvement in power factor (from 0.587 to 0.820) is a useful by-product of harmonic mitigation that also reduces utility demand charges.

The combined evidence of the time-domain waveforms, FFT spectra, RMS measurements and computed THD and power factor values therefore confirms that the proposed passive filter design successfully achieves all the objectives stated in Section 1.4.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

A single-phase AC railway traction power supply has been successfully modelled in MATLAB/Simulink, comprising a 50 Hz AC source, an equivalent overhead-line impedance, a step-down transformer, a single-phase full-bridge diode rectifier, a DC smoothing choke and a Maxon RE₆₅₃₈₈₉₈₉₇₀ permanent magnet DC traction motor. The harmonic distortion produced by this non-linear configuration was characterised in detail using FFT-based spectral analysis and quantified using the Total Harmonic Distortion (THD) and power factor metrics.

The unfiltered simulation revealed a highly distorted supply current with a THD of 62.93 % and a true power factor of only 0.587. The dominant harmonic content was confined to the 3rd, 5th and 7th harmonic orders, with magnitudes of 55.7 %, 20.2 % and 17.2 % of the fundamental, respectively. Three shunt single-tuned passive harmonic filters were designed using the standard resonance condition $f_r = 1 / (2\pi\sqrt{LC})$ and integrated on the AC side of the rectifier. The component values obtained were $R = 0.03 \Omega$, $L = 3.95 \text{ mH}$, $C = 285 \mu\text{F}$ for the 3rd-harmonic branch; $R = 0.03 \Omega$, $L = 5.8 \text{ mH}$, $C = 70 \mu\text{F}$ for the 5th-harmonic branch; and $R = 0.03 \Omega$, $L = 5.9 \text{ mH}$, $C = 35 \mu\text{F}$ for the 7th-harmonic branch.

With the filter bank in service, the supply current THD was reduced from 62.93 % to 7.65 %, comfortably below the IEEE 519 acceptability threshold of 8 %. The true power factor improved from 0.587 to 0.820, and the current waveform was restored to a near-sinusoidal shape. These results conclusively demonstrate that even a modest, three-branch passive filter bank is capable of delivering a major improvement in the power quality of an AC traction supply, while at the same time providing useful reactive-power compensation.

6.2 Future Scope

While the present work establishes the effectiveness of passive harmonic filtering in a representative single-phase traction circuit, several extensions are possible and would be valuable directions for further research:

Hardware prototype: A scaled-down laboratory prototype using a real PMDC motor, a single-phase bridge rectifier and discrete inductors and capacitors would validate the simulation results experimentally.

Detuned and high-pass filters: The addition of a high-pass damped filter to attenuate the 9th, 11th and higher-order harmonics would further reduce the THD towards the 5 % benchmark.

Hybrid active-passive filters: Combining the passive bank with a small active power filter would provide adaptive performance under changing traction loads while retaining most of the cost advantages of the passive solution.

Three-phase traction: Extension of the present analysis to three-phase 2×25 kV auto-transformer-fed traction systems would expand its applicability to high-speed railway corridors.

Regenerative braking: Incorporating a bidirectional converter and studying the harmonic behaviour during regeneration would address an increasingly important operational mode.

Real-load profiles: Driving the model with measured time-varying traction load profiles (start-up, cruise, coast, brake) would reveal the dynamic performance of the filters.

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